Structural Imagination Spaces: A Formal Theory of Layered Internal Structure

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20 August 2026

Abstract

The Theory of Structural Imagination (TSI) proposes that an internal representation for reasoning and action should be specified by the structures it preserves and the transformations it supports, rather than by its storage format alone. This paper gives a consolidated formal core. A state is a typed relational functor on a finite presented schema, a simplicial complex and metric on a common tagged carrier, a monotone filtration, a full-support mass, and action-conditioned tracked transitions. We define structural isomorphism and five-layer survivor preservation, including the cross-layer compatibility axiom, under which we prove a quantitative capacity bound on the relational defect. We also prove equivalence and composition properties, a six-component zero criterion including turnover, a hard-label metric–measure triangle inequality, and a quotient-level rollout bound under explicit equivariance and Lipschitz assumptions. These are conditional mathematical results about the specified representation class; they do not establish that human cognition or an unconstrained neural network implements it.

1 Introduction

Representation learning routinely stores internal states as vectors or tensors. That storage choice does not specify which entities persist, which relations compose, which metric quantities matter, how ordered and mass observables behave, or how an action changes the state. TSI begins with a narrower question: what mathematical data must be exposed if an internal state is intended to support structural simulation, reasoning, and action?

The formal object developed here is a finite presented schema together with a common tagged carrier, typed relations, a simplicial complex, a metric, a filtration, a full-support mass, and action-conditioned tracked transitions. These layers are allowed to interact, but they are not conflated. A vector may encode the data; its coordinates are not the complete structural specification.

The paper makes one integrated contribution through five components: (1) a state contract with a declared cross-layer compatibility axiom; (2) an exact structural isomorphism and its equivalence proof; (3) survivor-level preservation for five observables and its composition proof; (4) a six-defect zero criterion separating local preservation from turnover; and (5) categorical and quotient-level extensions, including metric–measure comparison and rollout stability.

The finite-relation category and presented-category descent are imported from the calculus of relations and categorical data formalisms [10, 16, 17]. Filtered persistence stability is imported from persistent homology [5, 6]; metric–measure couplings and gluing are imported from Gromov–Wasserstein and optimal-transport theory [14, 15]; and the rollout recurrence uses the standard Lipschitz error-propagation pattern [1]. The TSI-specific contribution is their integration under a common tagged state contract, the compatibility invariant, the label-preserving partial transition category, and the explicit separation of survivor preservation from temporal identity. The paper does not claim that these layers are necessary or sufficient for intelligence, that they describe human mathematical imagery, or that a neural model will discover them without inductive constraints.

2 Related Work and Positioning

Structured learning systems commonly impose object- and relation-centered inductive biases through interaction networks, graph networks, or structured world models [2, 3, 12]. These works specify learnable architectures and training objectives. TSI instead specifies a mathematical state contract, its equivalences, and conditional preservation properties. It neither supplies a neural discovery algorithm nor implies that the contract will emerge from an unconstrained learner.

Categorical database models treat a schema as a small category and an instance as a set-valued functor [13, 16]; finitely presented schemas can be given by directed multigraphs and path equations [17]. TSI imports this presentation discipline but uses finite binary relations as morphisms, following the standard calculus of relations [10]. Its additional simplicial, metric, filtration, mass, and tracked-transition layers are not part of those database formalisms.

The topological extension uses persistence diagrams and their stability theory [4–8, 20]. The metric–measure extension is a finite, hard-label-constrained variant of a Gromov–Wasserstein comparison [14]; it is distinct from formulations that jointly compare features and structure [18]. Standard coupling and gluing arguments are drawn from optimal transport [15, 19].

Bisimulation metrics for Markov decision processes build behavioral distances from rewards and transition kernels [9]. Multi-step model error has also been bounded under Lipschitz assumptions in Wasserstein geometry [1], while model-based policy optimization makes the practical case for restricting model-generated rollout horizons [11]. TSI does not identify its structural discrepancy with a bisimulation fixed point or a reward-sufficient abstraction. Its rollout result is the deterministic error recurrence that follows once a quotient-level extended pseudometric, equivariant updates, one-step errors, and Lipschitz constants have separately been supplied.

3 Formal Framework

3.1 Typed relational carriers

We use the category **FinRel** of finite sets and binary relations. Its objects are finite sets, and a morphism $R : X \nrightarrow Y$ is a subset of $X \times Y$. For $R : X \nrightarrow Y$ and $S : Y \nrightarrow Z$,

$$S \circ R = \{(x, z) : \exists y \in Y, (x, y) \in R,\ (y, z) \in S\}.$$

The identity on X is $\Delta_X = \{(x, x) : x \in X\}$.

Fix a finite quiver $Q = (Q_0, Q_1, s, t)$ and a finite set $\mathcal{E}$ of pairs of parallel paths in its free path category. The category $\mathrm{Path}(Q)$ has the vertices of Q as objects, finite composable paths as morphisms, length-zero paths as identities, and concatenation as composition. The schema is the presented category

$$\mathcal{S} = \mathrm{Path}(Q)/{\equiv_{\mathcal{E}}},$$

where $\equiv_{\mathcal{E}}$ is the smallest path congruence containing $\mathcal{E}$. The arrows of Q are the declared generators, and every schema morphism has a path representative subject to the declared equations. A typed relational carrier is a functor $F : \mathcal{S} \to \mathbf{FinRel}$ such that every $F(A)$ is nonempty and finite. A generator $e \in Q_1$ is interpreted as the relation $F([e])$. Functoriality requires $F(\mathrm{id}_A) = \Delta_{F(A)}$ and

$F(g \circ f) = F(g) \circ F(f)$. A label family is a collection of maps $\lambda_A : F(A) \to L_A$ into fixed typewise label sets.

Definition 3.1 (Integrated structural imagination state). *An integrated structural imagination state is a tuple*

$$\Sigma = (F, \lambda, K, d, f, \mu),$$

where F is a typed relational carrier, λ is a label family, K is a finite abstract simplicial complex whose vertex set is the tagged union $|F| = \coprod_A F(A)$, and d is a metric on $|F|$. The function $f : K \to \mathbb{R}$ is a monotone filtration and $\mu : |F| \to (0, 1]$ is a full-support probability mass function. The state also obeys the compatibility condition

$$(x, y) \in F([e]) \Longrightarrow \{x, y\} \in K \qquad (e \in Q_1).$$

The class of such states is denoted by $\mathsf{StrState}(Q, \mathcal{E})$.

The common tagged carrier is a scope decision. It permits relations, simplices, metric distances, filtration values, and mass to refer to the same typed entities. The compatibility condition is the minimal cross-layer constraint used in this paper: a declared generator relation must be visible as a one-simplex, but no converse or metric compatibility is assumed.

Definition 3.2 (Integrated structural isomorphism). *Let $\Sigma = (F, \lambda, K, d, f, \mu)$ and $\Sigma' = (G, \nu, L, \rho, g, \eta)$. An integrated structural isomorphism $\varphi : \Sigma \to \Sigma'$ is a family of bijections $\varphi_A : F(A) \to G(A)$ such that:*

1. *$\nu_A(\varphi_A x) = \lambda_A(x)$ for every A, x;*
2. *the tagged union $\bar{\varphi} : |F| \to |G|$ satisfies $\bar{\varphi}[K] = L$;*
3. *$\rho(\bar{\varphi} x, \bar{\varphi} y) = d(x, y)$ for all x, y;*
4. *$g(\bar{\varphi}\sigma) = f(\sigma)$ for every $\sigma \in K$;*
5. *$\eta(\bar{\varphi} x) = \mu(x)$ for every $x \in |F|$;*
6. *for every generator $e \in Q_1$,*

$$((\varphi_{s(e)} \times \varphi_{t(e)})[F([e])]) = G([e]).$$

We write $\Sigma \cong_{\mathrm{str}} \Sigma'$ when such a family exists.

Definition 3.3 (Tracked transition). *A tracked transition $p : \Sigma \rightharpoonup \Sigma'$ is a family $p = (p_A)_A$ in which each p_A is an injective map from a subset of $F(A)$ into $G(A)$ and*

$$\nu_A(p_A x) = \lambda_A(x) \qquad (x \in \operatorname{dom} p_A).$$

Its domain and image tagged unions are written D_p and I_p. The identity is the total identity family. If $p : \Sigma_0 \rightharpoonup \Sigma_1$ and $q : \Sigma_1 \rightharpoonup \Sigma_2$, their composite is the family $(q_A \circ p_A)|_{p_A^{-1}(\operatorname{dom} q_A)}$.

Definition 3.4 (Tagged extension of a partial transition). *For a tracked transition $p = (p_A)_A$, its tagged extension $\bar{p}$ is the map on $D_p = \coprod_A \operatorname{dom}(p_A)$ defined by $\bar{p}(x) = p_A(x)$ when $x \in F(A)$. Its image is $I_p = \coprod_A \operatorname{im}(p_A)$.*

Proposition 3.5 (The category of tracked transitions)**.** *Integrated states and tracked transitions form a category* $\mathsf{StrTrack}(Q, \mathcal{E})$*.*

Proof. The identity family is injective, label-preserving, and has the full carrier as its domain. A composite is injective on the restricted domain. If x is in its domain, label preservation follows successively from p and q. The domain of a triple composite is the set of elements on which all three partial maps are defined, and both parenthesizations have the same value there. Hence composition is associative. Restricting either composite by an identity leaves the same partial family, proving the unit laws. □

Definition 3.6 (Action-conditioned transition)**.** *Let* $\mathcal{A}$ *be an action set. An action-conditioned transition is a pair* (a, p) *with* $a \in \mathcal{A}$ *and* p *a tracked transition. The action label is an index on the transition, not additional data in the morphism of* $\mathsf{StrTrack}(Q, \mathcal{E})$*; consequently composition is defined on the tracked maps and does not concatenate incompatible single labels.*

4 Running Example: A Four-Surface Star State

We give one finite state that satisfies all state axioms and can be reused across the extensions. Let the presented schema have two objects B (block) and S (surface), one generator $e : B \to S$, and no nontrivial path equations. Set

$$F(B) = \{b\}, \qquad F(S) = \{s_1, s_2, s_3, s_4\},$$

and let $\lambda_B(b) = \mathsf{block}$ and $\lambda_S(s_i) = \mathsf{surface}$ for all i. Let $F([e]) = \{(b, s_i) : 1 \le i \le 4\}$. Let K be the star complex whose vertices are these five entities and whose nontrivial simplices are the four edges $\{b, s_i\}$. The compatibility condition holds by construction.

Define

$$d(x, x) = 0, \qquad d(b, s_i) = 1, \qquad d(s_i, s_j) = 2 \quad (i \ne j),$$

with symmetry understood. This is a metric: the only nontrivial triangle inequality is $2 \le 1 + 1$. Give every vertex filtration value 0, every edge filtration value 1, and every entity mass $1/5$. Call the resulting state $\Sigma_\star$. The filtration is monotone and the mass has full support.

Let $\Sigma'_\star$ be an isomorphic copy with primed entity names. The bijection $b \mapsto b'$ and $s_i \mapsto s'_i$ preserves labels, the star complex, distances, filtration values, masses, and the generator relation. Consequently,

$$d_B(\mathrm{Dgm}_k(f), \mathrm{Dgm}_k(g)) = 0 \quad (k \le r), \qquad \Delta_{\mathrm{mm},p}((|F|, d, \mu, \lambda), (|G|, \rho, \eta, \nu)) = 0 \quad (1 \le p < \infty).$$

The degree-zero persistence diagram has one point $(0, \infty)$ and four points $(0, 1)$, one for each component born before the four edges appear; there is no positive-degree class for this star filtration.

The same example also exposes the distinction between local preservation and temporal identity. Let p map b, s_1, s_2 to b', s'_1, s'_2 and leave s_3, s_4 unmatched. The survivor subcomplexes and restricted generator relations agree, all surviving distances, filtration values, and masses agree, so

$$m_K(p) = m_d(p) = m_R(p) = m_f(p) = m_\mu(p) = 0.$$

However,

$$b(p) = (4 - 2) + (4 - 2) = 4,$$

where the first term is source turnover in type S and the second is target turnover in type S. Thus zero layer defects do not certify full temporal identity. If instead all four surfaces and the block are matched by the primed copy, then all six quantities, including $b(p)$, vanish and the transition is a full structural isomorphism.

This example is deliberately finite and transparent. It does not claim that the star complex is a model of cognition; it verifies that the formal state class is nonempty and that the defect quantities distinguish survivor-level preservation from carrier turnover.

5 Layer Theorems

Proposition 5.1 (Structural equivalence). *The relation $\cong_{\mathrm{str}}$ on $\mathsf{StrState}(Q,\mathcal{E})$ is reflexive, symmetric, and transitive.*

Proof. The identity family preserves labels, simplices, distances, filtration values, masses, and generator relations, so reflexivity holds. If φ is an isomorphism, each φ_A^{-1} is a bijection; applying inverse maps to all six defining equalities gives the reverse conditions, so symmetry holds. If φ and ψ are composable, their typewise composites preserve labels. The simplex, metric, filtration, mass, and generator-relation equalities compose in sequence, giving all six conditions for $\psi\varphi$. Thus transitivity holds. $\square$

Proposition 5.2 (Generator preservation extends to paths). *If an integrated structural isomorphism preserves every generating relation, then it transports the relation assigned to every morphism of $\mathcal{S}$.*

Proof. For an identity morphism, a bijection transports the diagonal relation to a diagonal relation. Suppose the claim holds for paths $u : A \to B$ and $v : B \to C$. Membership in $F(v) \circ F(u)$ is witnessed by an intermediate $y \in F(B)$. The induction hypotheses transport both relations, and bijectivity of φ_B transports the witness to and from every element of $G(B)$. Hence the transported composite is $G(v) \circ G(u)$. Induction on path length, followed by the path equations, proves the claim for every schema morphism. $\square$

For a tracked transition p, write $K|_{D_p}$ for the induced subcomplex on surviving tagged entities and $F([e])|_{D_A,D_B}$ for the restriction of a generator relation to survivor domains.

Definition 5.3 (Survivor-level layer preservation). *A transition $p : \Sigma \rightharpoonup \Sigma'$ preserves the following layers on survivors when*

$$
\begin{aligned}
\bar{p}[K|_{D_p}] &= L|_{I_p} && \textit{(topology)},\\
d'(\bar{p}x, \bar{p}y) &= d(x,y) && \textit{for all } x,y \in D_p && \textit{(geometry)},\\
(p_A \times p_B)[F([e])|_{D_A,D_B}] &= G([e])|_{I_A,I_B} && \textit{for every generator } e : A \to B && \textit{(relations)},\\
g(\bar{p}\sigma) &= f(\sigma) && \textit{for every surviving simplex } \sigma && \textit{(filtration)},\\
\eta(\bar{p}x) &= \mu(x) && \textit{for every } x \in D_p && \textit{(mass)}.
\end{aligned}
$$

Theorem 5.4 (Composition closure). *Let $\Sigma_0 \xrightarrow{p} \Sigma_1 \xrightarrow{q} \Sigma_2$ be tracked transitions. If both transitions preserve any chosen subset of topology, geometry, relations, filtration, and mass on survivors, then $q \circ p$ preserves the same subset.*

Proof. For each type A, let $E_A = \{x \in \mathrm{dom}(p_A) : p_A x \in \mathrm{dom}(q_A)\}$, $J_A = p_A(E_A)$, and $H_A = q_A(J_A)$. These are the domains, intermediate survivors, and images of the composite. The first topology and filtration equalities restricted to $E = \coprod_A E_A$ give the structures on J, and the second equalities restricted to J give the structures on H; composing gives the required equalities. For geometry and mass, applying the two pointwise equalities successively gives $d_2(\bar{q}\bar{p}x, \bar{q}\bar{p}y) = d_0(x, y)$ and $\eta(\bar{q}\bar{p}x) = \mu(x)$. For a generator $e : A \to B$, restrict the first relational equality to $E_A \times E_B$, whose image is $J_A \times J_B$, and then restrict the second to $J_A \times J_B$. The product maps are bijections on the corresponding survivor products, so the relational equality composes. □

Corollary 5.5 (Trajectory preservation)**.** *Transitions preserving a chosen subset of these five layers form a wide subcategory of* $\mathsf{StrTrack}(Q, \mathcal{E})$*. Consequently, a finite trajectory whose edges preserve those layers has a composite transition preserving them.*

Proof. The identity transition preserves every layer. The composition theorem gives closure under composition, and induction on the number of trajectory edges gives the result. □

6 Persistent Topology Extension

This section adds a persistent topological observable to the integrated state. It is deliberately narrower than a claim that topology identifies structure: the carrier is a fixed finite simplicial complex, and the filtration is the only varying input in the stability result below.

Definition 6.1 (Filtered finite carrier)**.** *Let K be a finite abstract simplicial complex and let $\mathbb{F}$ be a field. A filtration function is a map $f : K \to \mathbb{R}$ satisfying $\sigma \subseteq \tau \Rightarrow f(\sigma) \leq f(\tau)$. Its sublevel complex at $a \in \mathbb{R}$ is*

$$K^f_a = \{\sigma \in K : f(\sigma) \leq a\}.$$

The persistent signature in degree k is the persistence diagram $\mathrm{Dgm}_k(f)$ of the homology persistence module $H_k(K^f_a; \mathbb{F})$, and the truncated signature is $\Pi_r(K, f) = (\mathrm{Dgm}_0(f), \ldots, \mathrm{Dgm}_d(f))$. This construction follows the standard finite persistence framework [8, 20].

The monotonicity condition makes every K^f_a a simplicial subcomplex: if τ is retained, every face σ is retained as well. Thus inclusion of sublevel complexes gives the persistence maps used in the definition.

Definition 6.2 (Persistent discrepancy)**.** *For two filtration functions f, g on the same finite carrier K, define*

$$\Delta_{\mathrm{pers},r}(f, g) = \max_{0 \leq k \leq r} d_B\big(\mathrm{Dgm}_k(f), \mathrm{Dgm}_k(g)\big),$$

where d_B is the bottleneck distance. The two filtrations are compared only after fixing the field, the carrier, the degree cutoff, and the units of the filtration values.

The following result is the standard stability theorem for persistence diagrams and persistence modules [5, 6].

Theorem 6.3 (Filtered stability; imported theorem)**.** *For a fixed finite simplicial complex K and monotone filtration functions $f, g : K \to \mathbb{R}$,*

$$d_B\big(\mathrm{Dgm}_k(f), \mathrm{Dgm}_k(g)\big) \leq \|f - g\|_\infty \quad \text{for every } k.$$

Consequently, $\Delta_{\mathrm{pers},r}(f, g) \leq \|f - g\|_\infty$.

Proof. Put $\delta = \|f - g\|_\infty$. For every simplex σ, the inequalities $g(\sigma) \leq f(\sigma) + \delta$ and $f(\sigma) \leq g(\sigma) + \delta$ hold. Therefore, for every a,

$$K^f_a \subseteq K^g_{a+\delta} \quad \text{and} \quad K^g_a \subseteq K^f_{a+\delta}.$$

The inclusions commute with the inclusions between sublevel complexes as a increases. Applying homology gives a δ-interleaving of the two persistence modules in degree k. Because K is finite, every vector space in either module is finite-dimensional; in particular, the modules are q-tame and admit interval decompositions and persistence diagrams [7]. The algebraic stability theorem [4, 5] gives a bottleneck distance no larger than the interleaving distance. Hence $d_B(\mathrm{Dgm}_k(f), \mathrm{Dgm}_k(g)) \leq \delta$. Taking the maximum over $0 \leq k \leq r$ proves the final inequality. The only non-elementary step is the imported algebraic stability theorem; the sublevel inclusions and interleaving construction are established here. □

Proposition 6.4 (Isomorphism invariance of the persistent signature)**.** *Let $\varphi : K \to L$ be a simplicial isomorphism and let $f : K \to \mathbb{R}$ and $g : L \to \mathbb{R}$ satisfy $g(\varphi(\sigma)) = f(\sigma)$ for every simplex σ. Then, for every k,*

$$d_B\big(\mathrm{Dgm}_k(f), \mathrm{Dgm}_k(g)\big) = 0.$$

Proof. Fix a. The value-preservation condition implies $\varphi(K_a^f) = L_a^g$. Since the restriction $\varphi_a : K_a^f \to L_a^g$ is a simplicial isomorphism, functoriality of simplicial homology gives $H_k(K_a^f; \mathbb{F}) \cong H_k(L_a^g; \mathbb{F})$. For $a \leq b$, the square formed by these restrictions and the inclusion-induced homology maps commutes. Thus the persistence modules are isomorphic and their persistence diagrams are identical, so the bottleneck distance is zero. □

Proposition 6.5 (The persistent signature is not complete for a labeled carrier)**.** *Even on one fixed labeled carrier, equality of all persistence diagrams need not determine the filtration as labeled structural data.*

Proof. Let K consist of two isolated vertices x, y with distinct labels. Set $f(x) = 0, f(y) = 1$ and $g(x) = 1, g(y) = 0$. Both degree-zero diagrams contain $(0, \infty)$ and $(1, \infty)$, and every positive-degree diagram is empty. Thus $\Pi_r(K, f) = \Pi_r(K, g)$ for every $r \geq 0$. However, the only label-preserving bijection fixes both vertices, and it does not preserve the filtration values. Hence the persistent signature does not determine the labeled filtered state. □

The extension certifies invariance and controlled sensitivity under its stated hypotheses. It does not establish completeness, comparison of unrelated carriers without a correspondence, or neural discovery of the carrier. Those questions require separate assumptions or empirical validation.

Proposition 6.6 (State-level persistent invariance)**.** *Let Σ and Σ' be integrated states and let $\varphi : \Sigma \to \Sigma'$ be a structural isomorphism. Then for every degree cutoff r,*

$$d_B(\mathrm{Dgm}_k(f), \mathrm{Dgm}_k(g)) = 0 \quad \textit{for every } k \leq r.$$

Proof. The filtration-preserving component of φ satisfies $g(\bar{\varphi}\sigma) = f(\sigma)$ for every simplex. Therefore the isomorphism-invariance proposition applies to the two filtered carriers. □

7 Metric–Measure Extension

The geometric layer can contain uncertain or repeated entities, so a set-level correspondence alone is not sufficient. We attach probability mass to the finite metric carrier and compare mass through couplings. The result below is an exactness theorem, not a finite-sample convergence theorem. Coupling and Gromov–Wasserstein background is standard [14, 15, 19].

Definition 7.1 (Full-support metric–measure state)**.** *A full-support metric–measure state is a tuple $M = (X, d, \mu, \lambda)$, where (X, d) is a finite metric space, $\mu : X \to (0, 1]$ satisfies $\sum_{x \in X} \mu(x) = 1$, and $\lambda : X \to \mathcal{L}$ is a label map. Full support means $\mu(x) > 0$ for every $x \in X$.*

Full support is required for the exactness result: a zero-mass entity could be added or moved without changing any coupling objective. The measure is part of the state, while coordinates or unlisted attributes are not implicitly treated as geometric evidence.

Definition 7.2 (Label-compatible coupling)**.** *For full-support states $M_X = (X, d_X, \mu_X, \lambda_X)$ and $M_Y = (Y, d_Y, \mu_Y, \lambda_Y)$, a coupling is a function $\pi : X \times Y \to \mathbb{R}_{\geq 0}$ with*

$$\sum_{y \in Y} \pi(x, y) = \mu_X(x), \qquad \sum_{x \in X} \pi(x, y) = \mu_Y(y).$$

It is label-compatible when $\pi(x, y) > 0 \Rightarrow \lambda_X(x) = \lambda_Y(y)$.

Definition 7.3 (p-metric–measure discrepancy)**.** *For $1 \leq p < \infty$ and a label-compatible coupling π, put*

$$\mathcal{D}_p(\pi)^p = \sum_{x, x' \in X} \sum_{y, y' \in Y} |d_X(x, x') - d_Y(y, y')|^p \pi(x, y) \pi(x', y').$$

The metric–measure discrepancy is

$$\Delta_{\mathrm{mm},p}(M_X, M_Y) = \inf_{\pi} \mathcal{D}_p(\pi),$$

where the infimum ranges over label-compatible couplings. If no such coupling exists, the discrepancy is $+\infty$.

This is a finite hard-label-constrained Gromov–Wasserstein-type objective [14]. It does not include coordinate costs, soft attribute costs, or a learned correspondence. Joint feature–structure costs, for example, lead to fused Gromov–Wasserstein formulations [18]; such additions define a different discrepancy and require their own identifiability analysis.

Theorem 7.4 (Zero metric–measure discrepancy is exact)**.** *Let M_X and M_Y be full-support metric–measure states. For every $1 \leq p < \infty$, $\Delta_{\mathrm{mm},p}(M_X, M_Y) = 0$ if and only if there is a bijection $\varphi : X \to Y$ such that, for every $x, x' \in X$,*

$$d_Y(\varphi(x), \varphi(x')) = d_X(x, x'), \qquad \lambda_Y(\varphi(x)) = \lambda_X(x), \qquad \mu_Y(\varphi(x)) = \mu_X(x).$$

Proof. Suppose first that such a bijection exists. Define

$$\pi_\varphi(x, y) = \begin{cases} \mu_X(x), & y = \varphi(x), \\ 0, & \text{otherwise.} \end{cases}$$

The first marginal is μ_X. For $y = \varphi(x)$, bijectivity and measure preservation make the second marginal equal to $\mu_X(x) = \mu_Y(y)$; hence this is a coupling. Label preservation makes it label-compatible. A positive product $\pi_\varphi(x, y)\pi_\varphi(x', y')$ can occur only with $y = \varphi(x)$ and $y' = \varphi(x')$, where the distance difference is zero. Thus $\mathcal{D}_p(\pi_\varphi) = 0$ and the discrepancy is zero.

Conversely, suppose the discrepancy is zero. The set of all couplings is a closed and bounded subset of the finite-dimensional space $\mathbb{R}^{X \times Y}$; imposing the finitely many label constraints keeps a closed subset. If it is nonempty, it is compact. Since $\mathcal{D}_p$ is continuous, a minimizing coupling π_* exists and satisfies $\mathcal{D}_p(\pi_*) = 0$.

Let $C = \{(x, y) : \pi_*(x, y) > 0\}$. Full support of each marginal implies that every x occurs in some pair in C and every y occurs in some pair in C, so C is a correspondence. For any two pairs $(x, y), (x', y') \in C$, their coefficient in the defining sum is positive. Every summand is nonnegative and the sum is zero, so $d_X(x, x') = d_Y(y, y')$.

If both (x, y) and (x, y') belong to C, applying this equality to the two pairs gives $d_Y(y, y') = d_X(x, x) = 0$, hence $y = y'$. If both (x, y) and (x', y) belong to C, the same argument gives $d_X(x, x') = 0$, hence $x = x'$. Therefore C is the graph of a bijection $\varphi : X \to Y$. The distance equality makes φ an isometry, and positive support makes it label-preserving. Finally, the first marginal gives $\mu_X(x) = \pi_*(x, \varphi(x))$, while the graph property makes the second marginal give $\mu_Y(\varphi(x)) = \pi_*(x, \varphi(x))$. Thus the bijection is measure-preserving. □

The theorem gives a precise zero-set interpretation for the metric–measure layer. It does not prove robustness under sampling, rates of empirical convergence, existence of a good coupling for arbitrary predicted carriers, or superiority over coordinate losses. These are separate empirical claims.

Proposition 7.5 (Label-compatible coupling feasibility). *A label-compatible coupling exists if and only if the total mass of every label agrees:*

$$\sum_{x:\lambda_X(x)=\ell} \mu_X(x) = \sum_{y:\lambda_Y(y)=\ell} \mu_Y(y) \qquad (\ell \in \mathcal{L}).$$

Proof. Necessity is the preceding marginal argument. For sufficiency, for each label ℓ choose the product coupling within its two label blocks, normalized by their common mass. Summing these blockwise couplings gives a nonnegative matrix with the required marginals and no cross-label mass. If a common label mass is zero, its block is omitted. □

Define the tagged label maps by $\bar{\lambda}(x) = (A, \lambda_A(x))$ for $x \in F(A)$ and $\bar{\nu}(y) = (A, \nu_A(y))$ for $y \in G(A)$.

Corollary 7.6 (State-level metric–measure invariance). *If $\Sigma \cong_{\mathrm{str}} \Sigma'$ then the induced full-support metric–measure states $(|F|, d, \mu, \bar{\lambda})$ and $(|G|, \rho, \eta, \bar{\nu})$ have zero metric–measure discrepancy for every $1 \le p < \infty$.*

Proof. Use the structural isomorphism as the measure-preserving, label-preserving isometry in the zero metric–measure discrepancy theorem. □

Proposition 7.7 (Extended pseudometric property of the hard-label discrepancy). *On full-support finite metric–measure states with tagged labels, $\Delta_{\mathrm{mm},1}$ is symmetric, vanishes on the diagonal, and satisfies the triangle inequality. It may take the value $+\infty$ when a label-compatible coupling does not exist.*

Proof. The identity-graph coupling gives diagonal vanishing. Transposing any label-compatible coupling preserves its marginals in reverse order and leaves $\mathcal{D}_1$ unchanged, which proves symmetry after taking infima. For the triangle inequality, let π_{XY} and π_{YZ} be label-compatible couplings. Full support gives $\mu_Y(y) > 0$, so

$$\gamma(x, y, z) = \frac{\pi_{XY}(x, y)\pi_{YZ}(y, z)}{\mu_Y(y)}$$

is well-defined. This is the standard gluing construction in optimal transport [15, 19]. Direct summation shows that its (X, Y) and (Y, Z) marginals are the original couplings. Let π_{XZ} be its

(X, Z) marginal. Whenever $\gamma(x, y, z) > 0$, label compatibility gives $\lambda_X(x) = \lambda_Y(y) = \lambda_Z(z)$; hence π_{XZ} is label-compatible. For two triples sampled from γ,

$$|d_X(x, x') - d_Z(z, z')| \leq |d_X(x, x') - d_Y(y, y')| + |d_Y(y, y') - d_Z(z, z')|.$$

Summing this inequality against $\gamma \otimes \gamma$ yields $\mathcal{D}_1(\pi_{XZ}) \leq \mathcal{D}_1(\pi_{XY}) + \mathcal{D}_1(\pi_{YZ})$. Taking infima over the two original couplings proves the triangle inequality. If either coupling is infeasible, the corresponding extended value is $+\infty$ and the inequality is immediate. □

8 Categorical Descent Extension

The relational carrier in the formal framework is a functor on a schema category. This section makes the descent condition explicit when the schema is given by generators and path equations. The purpose is to prevent a list of primitive relations from being mistaken for a well-defined categorical state. Schemas as categories and graph-and-equation presentations are standard in functorial data models [16, 17]; TSI changes the instance codomain from sets and functions to finite sets and binary relations.

Definition 8.1 (Generator realization). *A generator realization assigns a finite nonempty set X_A to every $A \in Q_0$ and a relation $R_e \subseteq X_{s(e)} \times X_{t(e)}$ to every $e \in Q_1$. Its free relational extension is*

$$\widetilde{F}(1_A) = \Delta_{X_A}, \qquad \widetilde{F}(e_n \cdots e_1) = R_{e_n} \circ \cdots \circ R_{e_1}.$$

Proposition 8.2 (The finite-relation category). *Finite sets, binary relations, relational composition, and diagonal relations form a category. In particular, the path formula in the preceding definition is independent of parenthesization. This is the standard category of relations [10]; the elementary verification is included for completeness.*

Proof. Let $R : X \nrightarrow Y$, $S : Y \nrightarrow Z$, and $T : Z \nrightarrow W$. A pair (x, w) belongs to $T \circ (S \circ R)$ exactly when there are $y \in Y$ and $z \in Z$ with $(x, y) \in R$, $(y, z) \in S$, and $(z, w) \in T$. Expanding $(T \circ S) \circ R$ gives exactly the same existential condition, so composition is associative. A pair belongs to $\Delta_Y \circ R$ exactly when its intermediate element is forced by the diagonal to equal the target; hence $\Delta_Y \circ R = R$. The analogous argument gives $R \circ \Delta_X = R$. These are the category axioms. □

Theorem 8.3 (Descent through path equations). *A generator realization descends to a functor $F : \mathcal{S} \to$ **FinRel** satisfying $F([e]) = R_e$ if and only if*

$$\widetilde{F}(p) = \widetilde{F}(q) \qquad \textit{for every } (p, q) \in \mathcal{E}.$$

When it exists, the descended functor is unique.

Proof. Assume first that a descended functor F exists and satisfies $F([e]) = R_e$. Let $\pi : \mathrm{Path}(Q) \to \mathcal{S}$ map each path to its equivalence class. We first verify the factorization from the generator data. For a length-zero path, functoriality gives $F([1_A]) = \Delta_{X_A} = \widetilde{F}(1_A)$. If $p = e_n \cdots e_1$ and the equality holds for paths of length $n - 1$, then functoriality and $F([e_i]) = R_{e_i}$ give

$$F([p]) = F([e_n]) \circ F([e_{n-1} \cdots e_1]) = R_{e_n} \circ \widetilde{F}(e_{n-1} \cdots e_1) = \widetilde{F}(p).$$

Thus $\widetilde{F} = F \circ \pi$ by induction on path length. If $(p, q) \in \mathcal{E}$, then $\pi(p) = \pi(q)$ by construction. Therefore $\widetilde{F}(p) = F(\pi(p)) = F(\pi(q)) = \widetilde{F}(q)$.

Conversely, assume equality for every declared equation. Define $p \sim q$ exactly when $\widetilde{F}(p) = \widetilde{F}(q)$, for parallel paths. Equality is reflexive, symmetric, and transitive. If $p \sim q$ and a, b are compatibly composable paths, functoriality of the free extension and associativity give

$$\widetilde{F}(bpa) = \widetilde{F}(b) \circ \widetilde{F}(p) \circ \widetilde{F}(a) = \widetilde{F}(b) \circ \widetilde{F}(q) \circ \widetilde{F}(a) = \widetilde{F}(bqa).$$

Thus $\sim$ is a path congruence containing $\mathcal{E}$. By minimality, $\equiv_{\mathcal{E}} \subseteq \sim$. Hence $F([p]) = \widetilde{F}(p)$ is well-defined. The identity and composition laws are inherited from $\widetilde{F}$, so F is a functor and $\widetilde{F} = F \circ \pi$. Every morphism of $\mathcal{S}$ has a path representative, so any functor satisfying the factorization is forced to have $F([p]) = \widetilde{F}(p)$; uniqueness follows. $\square$

Proposition 8.4 (Generator criterion for natural equivalence)**.** *Let $F, G : \mathcal{S} \to$ **FinRel** be functors with the same object carriers and let $\varphi_A : F(A) \to G(A)$ be bijections. The graphs $\mathrm{Gr}(\varphi_A)$ form a natural isomorphism if and only if, for every generator $e : A \to B$ and every x, y,*

$$(x, y) \in F([e]) \iff (\varphi_A(x), \varphi_B(y)) \in G([e]).$$

Proof. If the family is natural, its naturality square at each generator is $G([e]) \circ \mathrm{Gr}(\varphi_A) = \mathrm{Gr}(\varphi_B) \circ F([e])$. Expanding graph composition gives the displayed biconditional: one direction follows by transporting a related pair, and the other uses surjectivity of φ_B.

Conversely, the biconditional transports each generator relation. It also holds for identities because a bijection transports a diagonal relation to a diagonal relation. For composable paths, membership in a relational composite is witnessed by an intermediate entity; applying the criterion to both factors and using bijectivity transports that witness. Induction on path length gives the criterion for every path. Since both functors satisfy the same path equations, it is independent of the chosen representative. The corresponding naturality equality therefore holds for every schema morphism. Each graph is an isomorphism in **FinRel**, so the natural transformation is a natural isomorphism. $\square$

If a declared path equation is not satisfied by the generator relations, the assignment remains a valid functor on the free path category but is not a state over the presented schema. Thus categorical validity is an additional mathematical gate, not an interpretive label attached after prediction. The extension does not assert that a learner discovers the correct schema or that generator-level agreement is sufficient without checking the equations.

9 Dynamics and Temporal Identity Extension

A structural transition is a partial identity claim: it says which typed entities persist across an update. This section makes finite action histories and turnover-aware defects explicit. Endpoint agreement alone is not treated as evidence for temporal identity.

Definition 9.1 (Finite action-prefix category)**.** *Let $\mathcal{A}$ be a nonempty finite action alphabet and let $H \geq 0$. The category $\mathcal{H}^H_{\mathcal{A}}$ has as objects all words $u \in \mathcal{A}^*$ with $|u| \leq H$. There is one morphism $u \to v$ exactly when u is a prefix of v; its identity is the empty extension and composition concatenates suffixes.*

The prefix relation is a finite poset category. A morphism $u \to v$ represents executing the suffix of v that remains after u.

Definition 9.2 (Tracked action history)**.** *A tracked action history of horizon H is a functor $\mathcal{D} : \mathcal{H}_{\mathcal{A}}^{H} \to \mathsf{StrTrack}(Q, \mathcal{E})$. Thus $\mathcal{D}(u)$ is the state at action word u, while $\mathcal{D}(u \to v)$ is the composite tracked transition from the prefix state to the descendant state.*

Theorem 9.3 (Unique extension from one-step action edges)**.** *Suppose a state Σ_u is assigned to every word with $|u| \le H$ and a tracked transition $p_{u,a} : \Sigma_u \rightharpoonup \Sigma_{ua}$ is assigned whenever $|u| < H$. There is a unique functor $\mathcal{D}$ taking these one-step transitions as prescribed values.*

Proof. Set $\mathcal{D}(u) = \Sigma_u$ and map every identity to the identity transition. If $v = ua_1 \cdots a_k$, define

$$\mathcal{D}(u \to v) = p_{ua_1 \cdots a_{k-1}, a_k} \circ \cdots \circ p_{u,a_1}.$$

Associativity gives independence of parenthesization, and splitting an edge list at any intermediate prefix gives functoriality. The empty list gives identities. Any functor with the prescribed one-step values must send every longer prefix morphism to this composite, proving uniqueness. ☐

Theorem 9.4 (Trajectory preservation along action histories)**.** *Let $u \preceq v$ in a tracked action history. If every one-step edge on the path from u to v preserves any chosen subset of topology, geometry, relations, filtration, and mass on survivors, then $\mathcal{D}(u \to v)$ preserves the same subset.*

Proof. The unique extension theorem expresses $\mathcal{D}(u \to v)$ as the composite of the one-step edges. The five-layer composition theorem applies successively. ☐

Definition 9.5 (Turnover and layer defects)**.** *For $p : \Sigma \rightharpoonup \Sigma'$ with $\Sigma = (F, \lambda, K, d, f, \mu)$ and $\Sigma' = (G, \nu, L, \rho, g, \eta)$, let*

$$b(p) = \sum_{A} (|F(A)| - |\operatorname{dom}(p_A)| + |G(A)| - |\operatorname{im}(p_A)|).$$

Define

$$\begin{aligned}
m_K(p) &= |\bar{p}[K|_{D_p}] \mathbin{\triangle} L|_{I_p}|, \\
m_d(p) &= \max_{x,y \in D_p} |d(x,y) - \rho(\bar{p}x, \bar{p}y)|, \\
m_R(p) &= \sum_{e \in Q_1} |(p_{s(e)} \times p_{t(e)})[F([e])|_{D_{s(e)}, D_{t(e)}}] \mathbin{\triangle} G([e])|_{I_{s(e)}, I_{t(e)}}|, \\
m_f(p) &= \max_{\sigma \in K|_{D_p}} \begin{cases} |f(\sigma) - g(\bar{p}\sigma)|, & \bar{p}\sigma \in L, \\ +\infty, & \bar{p}\sigma \notin L, \end{cases} \\
m_\mu(p) &= \sum_{x \in D_p} |\mu(x) - \eta(\bar{p}x)|.
\end{aligned}$$

If D_p is empty, set $m_d(p) = m_f(p) = m_\mu(p) = 0$. Here $D_A = \operatorname{dom}(p_A)$ and $I_A = \operatorname{im}(p_A)$. The $+\infty$ convention makes the filtration defect defined even when a surviving source simplex is not mapped to a target simplex.

For $e : A \to B$ and tagged subsets $D_A \subseteq F(A)$ and $D_B \subseteq F(B)$, define the ordered simplicial-pair capacity

$$c_e(K; D_A, D_B) = \big|\{(x,y) \in D_A \times D_B : \{x,y\} \in K\}\big|.$$

This convention includes a permitted loop (x,x) through the vertex $\{x\} \in K$ and therefore does not silently assume irreflexive relations.

Proposition 9.6 (Compatibility bounds the relational defect)**.** *For every tracked transition $p : \Sigma \rightharpoonup \Sigma'$ between compatible states,*

$$m_R(p) \le \sum_{e \in Q_1} \big(c_e(K; D_{s(e)}, D_{t(e)}) + c_e(L; I_{s(e)}, I_{t(e)})\big).$$

Without the compatibility axiom, this bound need not hold. The bound is a capacity bound and is not claimed to be tight.

Proof. For each generator e, compatibility puts the restricted source relation inside the source simplicial-pair set counted by $c_e(K; D_{s(e)}, D_{t(e)})$. Injectivity of $p_{s(e)} \times p_{t(e)}$ preserves its cardinality. Target compatibility likewise puts the restricted target relation inside the set counted by $c_e(L; I_{s(e)}, I_{t(e)})$. The cardinality of a symmetric difference is at most the sum of the cardinalities of its two operands; summing over generators proves the bound.

For an explicit failure without compatibility, let Q have distinct objects A, B and one generator $e : A \to B$. Take $F(A) = \{x\}$, $F(B) = \{y\}$, $G(A) = \{x'\}$, and $G(B) = \{y'\}$, with the displayed elements carrying their respective distinct type tags. Let K and L contain only their vertices, set $F([e]) = \{(x, y)\}$ and $G([e]) = \varnothing$, and let the total tracked transition map $x \mapsto x'$ and $y \mapsto y'$. Then both capacities are zero, while

$$m_R(p) = |\{(x', y')\} \triangle \varnothing| = 1.$$

Thus the bound can fail when the source compatibility axiom is removed. □

Proposition 9.7 (Zero criterion for a tracked transition)**.** *For every tracked transition p, $m_K(p) = 0$, $m_d(p) = 0$, $m_R(p) = 0$, $m_f(p) = 0$, and $m_\mu(p) = 0$ are respectively equivalent to preservation of topology, geometry, generator relations, filtration, and mass on survivors. Also, $b(p) = 0$ is equivalent to every component being a total bijection. Consequently, all six quantities vanish exactly when p is a full integrated structural isomorphism.*

Proof. All five layer statements follow because finite symmetric differences, maxima of nonnegative terms, and sums of nonnegative terms vanish exactly when all their component conditions hold. The $+\infty$ case ensures that $m_f(p) = 0$ implies that every surviving simplex has a simplex image. For $b(p)$, vanishing forces both the domain and image of every component to equal the full finite carrier; injectivity then gives a total bijection. Combining these facts with the label-preserving definition of tracked transition gives all six conditions of integrated structural isomorphism. The converse follows by substitution. □

Proposition 9.8 (Survivor preservation can be vacuous)**.** *For any two nonempty integrated states, the transition with empty domain and empty image preserves topology, geometry, relations, filtration, and mass on survivors, while having strictly positive turnover.*

Proof. All induced survivor structures and all quantified comparisons are empty, while nonempty endpoint carriers make the turnover sum positive. □

The dynamics extension therefore separates local structural preservation, temporal identity, and turnover. It does not claim that terminal state correctness identifies the tracking map, that a formal action branch is a causal counterfactual, or that a learner discovers the correct update mechanism.

10 Dynamics and Rollout Stability

Static structure alone does not make a transition model well-defined. We require updates to respect structural equivalence before measuring their error.

Bisimulation metrics construct task-dependent behavioral distances for Markov decision processes from reward and transition information [9]. Lipschitz model analyses propagate one-step error over multiple steps in Wasserstein geometry [1]. The discrepancy below is neither asserted to be a bisimulation fixed point nor derived from rewards. The rollout theorem is instead conditional on a supplied structural pseudometric and supplied Lipschitz constants.

Definition 10.1 (Extended structural pseudometric)**.** *Let $\mathcal{X} = \mathsf{StrState}(Q, \mathcal{E})/\cong_{\mathrm{str}}$. An extended structural pseudometric on a subset $\mathcal{X}_0 \subseteq \mathcal{X}$ is a map*

$$\Delta_{\mathrm{str}} : \mathcal{X}_0 \times \mathcal{X}_0 \to [0, \infty]$$

that is symmetric, vanishes on the diagonal, satisfies the triangle inequality, and is allowed to take the value $+\infty$. A finite-distance component is a subset on which all pairwise values are finite. The rollout theorem below is conditional on these properties; it does not claim that any particular layer discrepancy automatically has them.

Proposition 10.2 (Existence of an admissible discrepancy)**.** *For every subset $\mathcal{X}_0$, the discrete discrepancy*

$$\Delta^{\mathrm{disc}}_{\mathrm{str}}(x, y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y, \end{cases}$$

is a structural metric on $\mathcal{X}_0$. Every map $T : \mathcal{X}_0 \to \mathcal{X}_0$ is 1-Lipschitz under this metric.

Proof. Symmetry and diagonal vanishing are immediate. For the triangle inequality, if $x = z$ the result is trivial; otherwise at least one of $x \neq y$ or $y \neq z$ holds, so $\Delta^{\mathrm{disc}}_{\mathrm{str}}(x, z) = 1 \leq \Delta^{\mathrm{disc}}_{\mathrm{str}}(x, y) + \Delta^{\mathrm{disc}}_{\mathrm{str}}(y, z)$. A map cannot increase a zero distance, and the only positive input distance is 1, so the Lipschitz claim follows. □

The discrete example is intentionally degenerate for rollout: if the one-step tolerance is less than 1, the predicted and true classes must coincide; if it is at least 1, the resulting bound can be vacuous because the diameter is 1.

The metric–measure result gives a non-discrete admissible instance. For a state Σ, let $M(\Sigma)$ be its induced full-support metric–measure state with tagged labels, and define

$$\Delta^{\mathrm{mm}}_{\mathrm{str}}([\Sigma], [\Sigma']) = \Delta_{\mathrm{mm},1}(M(\Sigma), M(\Sigma')).$$

Structural isomorphisms induce label- and measure-preserving isometries, so the value is independent of the chosen representatives. Symmetry, diagonal vanishing, and the triangle inequality make this an extended structural pseudometric on $\mathcal{X}$. It is not a metric there: zero value identifies the metric–measure layer but need not identify the simplicial complex or the generator relations.

Assumption 10.3 (Quotient-defined Lipschitz action models)**.** *Let $\mathcal{X}_0$ be a finite-distance component invariant under true and predicted updates. For each action a, assume that the updates $T_a, \widehat{T}_a : \mathcal{X}_0 \to \mathcal{X}_0$ are well-defined, that T_a is L_a-Lipschitz under an extended structural pseudometric Δ_{str}, and that $\Delta_{\mathrm{str}}(\widehat{T}_a z, T_a z) \leq \varepsilon_a$ at every predicted-rollout state z where the estimate is invoked.*

The well-definedness condition is an equivariance assumption: isomorphic input representatives must produce isomorphic outputs. It does not follow from the state definition. A pseudometric is sufficient for the recurrence; a genuine metric is required only when distinct quotient classes must be separated.

Theorem 10.4 (Action-conditioned rollout bound)**.** *For actions $a_1, \ldots, a_n$ and common initial class x_0, define $x_i = T_{a_i} x_{i-1}$ and $\widehat{x}_i = \widehat{T}_{a_i} \widehat{x}_{i-1}$. Under the assumption above,*

$$\Delta_{\mathrm{str}}(\widehat{x}_n, x_n) \leq \sum_{i=1}^{n} \varepsilon_{a_i} \prod_{j=i+1}^{n} L_{a_j},$$

where an empty product equals 1.

Proof. Let $e_i = \Delta_{\mathrm{str}}(\widehat{x}_i, x_i)$. The triangle inequality gives

$$e_i \leq \Delta_{\mathrm{str}}(\widehat{T}_{a_i} \widehat{x}_{i-1}, T_{a_i} \widehat{x}_{i-1}) + \Delta_{\mathrm{str}}(T_{a_i} \widehat{x}_{i-1}, T_{a_i} x_{i-1}) \leq \varepsilon_{a_i} + L_{a_i} e_{i-1}.$$

Since $e_0 = 0$, induction on i expands this recurrence into the displayed sum. Each induction step multiplies all previous terms by L_{a_i} and adds ε_{a_i}, which is exactly the claimed product pattern. □

Proposition 10.5 (No uniform rollout bound without uniform Lipschitz control)**.** *A common one-step error bound does not imply a nontrivial multi-step bound uniform over a family of models whose Lipschitz constants are unbounded.*

Proof. Take the metric component $[0, 1]$ with discrepancy $\Delta_{\mathrm{str}}(u, v) = c|u - v|$. For $\eta > 0$, let $T_M(u) = \min\{1, Mu\}$ and $\widehat{T}_M(u) = \min\{1, Mu+\eta\}$. Both maps are well-defined on this component, and T_M is M-Lipschitz. The one-step discrepancy is at most $c\eta$, independently of M. Starting at zero, choose M so that $M\eta+\eta \geq 1$; the predicted state reaches one at step two while the true state remains zero. Thus a fixed two-step error survives as $\eta \to 0$ when no common Lipschitz bound is imposed. □

Proposition 10.6 (Quotient equivariance is independently necessary)**.** *A representative-level update that maps isomorphic input states to non-isomorphic output states does not induce a well-defined update on $\mathcal{X} = \mathsf{StrState}(Q, \mathcal{E}) / \cong_{\mathrm{str}}$.*

Proof. Let $\Sigma \cong_{\mathrm{str}} \Sigma'$ and suppose a representative-level rule sends them to Θ and Θ' with $\Theta \ncong_{\mathrm{str}} \Theta'$. The quotient class of the input is the same whether represented by Σ or Σ', but the proposed output class differs. Therefore no function on input equivalence classes can agree with this rule on both representatives. □

11 Interpretation and Limits

The formal state has five coupled structural observables and a temporal transition layer. Topology concerns qualitative organization such as adjacency and connectedness. Geometry concerns metric information. Relations concern typed entities, generator morphisms, and composition. Filtration and mass add ordered and measure-theoretic observables on the same carrier. Dynamics concerns action-conditioned tracked transitions, histories, turnover, interventions, and rollouts. The common tagged carrier and the compatibility axiom couple these layers without identifying them.

The formal results establish conditional statements about explicitly specified objects. They do not prove that these observables are necessary or sufficient for intelligence, that human cognition literally uses this state object, or that an unconstrained neural network will discover it. They also do not prove that a structurally specified model is universally superior. Those statements require separate cognitive and empirical validation.

The principal design boundary is therefore deliberate: a structural claim must name its carrier, equivalence, observables, transformations, and discrepancy before it can be evaluated. A scalar predictive loss may be useful, but it cannot by itself determine structural equivalence or transition preservation. Conversely, a structural discrepancy is meaningful only after its task-relevant alignment and invariants are specified.

The formalism takes the entity set, its typing, and its cross-time identity as given data of the state. It does not supply a procedure for discovering entities from raw observation or for maintaining identity under perception error. Those stages precede everything specified here.

The companion empirical papers test bounded consequences of this framework on finite mechanism families. Those experiments should be read as validation of the stated consequences, not as validation of the entire theory or of real-world cognition.

12 Conclusion

This paper turns the TSI proposal into a formal core. An integrated structural state is a typed relational functor on a finite presented schema, a simplicial complex and metric on a common tagged carrier, a monotone filtration, a full-support mass, and action-conditioned tracked transitions. The compatibility axiom makes generator relation support a subcomplex of the simplicial one-skeleton. Structural equivalence is an equivalence relation; preservation of the five observables composes on survivors; the six defects characterize full structural isomorphism together with turnover; and the quotient-level rollout bound makes equivariance and stability assumptions explicit.

These theorems do not settle the cognitive or machine-learning hypotheses that motivated TSI. They provide definitions, proofs, and failure boundaries needed to state those hypotheses without leaving intuition as an untestable remainder.

The next decisive comparison is not a factorized head versus a generic sparse head. It contrasts two models that have both recovered the same dependency graph under entity creation and deletion, identity maintenance, and relation reconfiguration. For that comparison to identify anything, both arms must receive identical observable information and differ only in representational commitment. Supplying identity or typed relations as input to one arm only would make the result an artifact of information asymmetry, the same class of confound diagnosed in the original finite-family audit. The turnover quantity $b(p)$, its zero criterion, and the vacuity proposition for survivor-only preservation are already formalized here; a turnover-focused study is the smallest self-contained instance of this design.

Acknowledgments

We thank the colleagues and acquaintances who reviewed the ideas underlying this study.

Data and Code Availability

This theoretical study introduces no new empirical dataset. The bilingual manuscript sources and reference implementation are available at `https://github.com/nowtast/tsi`.

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